Form 37 game. 28 Jan 2027 Quotient equation $\frac{A(x)}{B(x)} = k$

1. Warming · Flash [EQ] For what value of x The expression $\frac{1}{x-3}$ Isn't it defined?

Rep. $x = 3$.

2. Heart · Method [EQ] Solve $\frac{2x-1}{x+3} = 0$.

Rep. $x \neq -3$ and $2x - 1 = 0 : x = \frac{1}{2}$.

3. Heart + · Method [EQ] Solve $\frac{x+1}{x-2} = 3$.

Rep. $x \neq 2\,;\, x + 1 = 3(x-2)$either $x = \frac{7}{2}$. Control: $4.5_{\overline{1,5}} = 3$.

4. Back · Develop [DEV] Develop and reduce $(x+2)(x-5)$.

Rep. $x^2 - 3x - 10$.

5. Red thread · Code reading [ALG] What Returned $g(3)$ and $g(-5)$?

```
def g(x):
    if x > 0:
        returns 2*x
    Else:
        Return -x
```

Rep. $g(3) = 6$ and $g(-5) = 5$.

Condition evaluated prior to calculation: branch depends on the sign of x.

Form 38 Mon 1st Feb. 2027 Sign of a product or quotient; signboards

1. Warming · Flash [EQ] Sign of 2 $x - 4$?

Rep. Negative for $x < 2$, zero in 2, positive for $x > 2$.

2. Heart · Method [EQ] Draw up the signboard of $(x-1)(x+3)$.

Rep. Roots -3 and 1: positive on $]-\infty;-3[$negative on $]-3;1[$positive on $]1;+\infty[$.

Product sign: column-by-column sign rule.

3. Heart + · Method [EQ] Solve $\frac{x+2}{x-1} \geqslant 0$.

Rep. Positive numerator for $x \geqslant -2$, positive denominator for $x > 1\,;\, S =]-\infty;-2] \cup]1;+\infty[$ (1 is Excluded).

4. Back · Flash [PRB] 32 card game: probability of pulling a heart ? a king ?

Rep. $\frac{8}{32} = \frac{1}{4}\,;\, \frac{4}{32} = \frac{1}{8}$.

5. Red thread · Refuter [LIT] Refute: "for all real x, $x^2 \geqslant x$ ».

Rep. Counter-example: $x = \frac{1}{-} : x^2 = \frac{1}{-} < \frac{1}{-} = x$.

A single counter-example is enough to refute a universal proposal.

Proportions and trends

Form 39 Mar. 2 Feb. 2027 Proportions and percentages; proportion

1. Warming · Flash [PRO] Calculate 15% of 240.

Rep. $0{,}15 \times 240 = 36$.

2. Heart · Method [PRO] 30% of students are interns and 40% of interns practice dance.
proportion of students are interns who dance?

Rep. $0{,}3 \times 0{,}4 = 0{,}12 = 12\%$: proportion (multiply)

3. Heart + · Problem [PRO] A high school has 800 students, 55 per cent of whom are girls; 20 per cent of girls are Latin.
How many Latinist girls?

Rep. $800 \times 0{,}55 = 440$ girls; $440 \times 0{,}20 = 88$.

4. Back · Flash [QP] 3 notebooks cost 4.50 €. Price of 7 notebooks?

Rep. $1{,}50 \times 7 = 10{,}50$ €.

5. Red thread · True/false [EQ] True or false? Justify. -2 is solution of equation $3x + 1 = -5$.

Rep. True. $3 \times (-2) + 1 = -6 + 1 = -5$.

Form 40 Game. 4 Feb. 2027 Absolute change, multiplier coefficient, rate of change

1. Warming · Flash [PRO] A price goes from 80 Ar to 92 Ar. What is the absolute variation?

Rep. $92 - 80 = 12$ Ar.

2. Heart · Method [PRO] For the same evolution, give the rate of change and the multiplier coefficient.

Rep. $t = \dfrac{12}{80} = 0{,}15 = 15\,\%$; $CM = 1 + t = 1{,}15$.

3. Heart + · MCQ [PRO] A 20% decrease corresponds to the multiplier coefficient:

A. 0,2 **B.** $-0{,}8$ **C.** 1,2 **D.** 0,8

Rep. D. 0,8

$CM = 1 + t = 1 - 0{,}20$.

4. Back · Flash [STA] Temperatures -2, 0, 3, 5, 9 (in °C).

Rep. Average 3 °C; extent $9 - (-2) = 11$ °C.

5. Red thread · Flash [SOL] Calculate the volume and total area of a 3 cm edge cube.

Rep. $V = 3^3 = 27\ \text{cm}^3$; $A = 6 \times 3^2 = 54\ \text{cm}^2$.

Form 41 Mon 8 Feb 2027 Successive developments

1. Warming · Flash [CAL] Calculate $1.2 \times 0{,}9$.

Rep. 1,08.

2. Heart · Method [PRO] A 20% increase is followed by a 10% decrease. What is the overall rate?

Rep. $1{,}2 \times 0{,}9 = 1.08$: overall increase of 8%.

Rates do not add up; coefficients are multiplying.

3. Heart + · True/false [PRO] True or false? Justify. A 50% increase followed by a 50% decrease brings back the initial value.

Rep. Wrong. $1.5 \times 0{,}5 = 0.75$: overall decrease of 25%.

4. Back · Flash [SUB] Calculate $x^2 - x$ for $x = \dfrac{1}{2}$.

Rep. $-\dfrac{1}{4}$.

5. Red thread · Write [DEM] $ABCD$ is a parallelogram such as $AC = BD$. What is its nature? Justify in one sentence.

Rep. Rectangle: a parallelogram whose diagonals are even long is a rectangle.

Form 42 Mar. 9 Feb. 2027 Converse development; proportion and evolution

1. Warming · Flash [CAL] Calculate $\dfrac{1}{1{,}25}$.

Rep. 0,8.

2. Heart · Method [PRO] After an increase of 25%, what drop brings down to the initial value?

Rep. $CM_{\text{rec}} = \dfrac{1}{1{,}25} = 0.8$: decrease by 20% (not 25%).

Trap: believe that an increase of 25 % offset by a decrease of 25 %.

3. Heart + · Interpret [PRO] Of 250 candidates, 150 are admitted; last year there were 200 candidates. Calculate the proportion of admissions and then the rate of change in the number of candidates.